Wheeler Memory

A Memory System That Remembers Like You Do

Darman: Cellular automata meet associative recall.

Why Do You Remember Your First Kiss Differently Every Time?

Context shapes memory. The same event, reconstructed through different lenses.

Same memory, infinite recalls.

That's human memory.

The Problem With Today's AI

ChatGPT has no memory between conversations.

- Never forgets
- Never uncertain
- Never hedges

But humans do all three.

They confabulate confidently. They should say "I'm not sure" and mean it.

What If AI Could Actually Remember?

Forget things? ✓

Hedge on uncertain memories? ✓

Reconstruct context-dependently? ✓

That's Wheeler Memory.

Cellular Automata: Inspired by Conway, Solving Memory

64×64 grid. Three simple rules per cell:

- **Peak** (local max) → push +1
- **Valley** (local min) → push −1
- **Slope** → flow uphill

The grid evolves. Chaos settles into stable patterns. Those patterns ARE memories.

How It Works: Storing a Memory

1. Text enters the system
2. Text is hashed (SHA-256)
3. CA evolves for ~40–100 ticks
4. Chaos converges to a stable pattern (attractor)
5. Pattern is stored as numbers on disk

In ~3 milliseconds. GPU: 10x faster.

How It Works: Recalling a Memory

1. You ask a question (query)
2. System finds the closest stored memory (Pearson correlation)
3. Blends it with your query context (30% query, 70% stored)
4. Re-evolves the blend through CA
5. Out comes a reconstructed answer

Context colors the recall.

The Magic: Reconstruction

Same stored memory.

Different query contexts.

Different reconstructed outputs.

Same memory. Infinite variations. Like your brain.

Temperature: Forgetting Built In

Memories have a lifespan:

- **New** (cool) — "I vaguely remember..."
- **Warm** (frequently used) — "I think..."
- **Hot** (recent) — "I distinctly remember..."

Half-life: 7 days without recall.

Decay matters. Age matters.

Epistemic Humility: The LLM Mirrors Memory Quality

Hot memory (≥0.6 temperature):

> "I remember clearly that..."

Warm memory (≥0.3):

> "I believe... but I'm not entirely certain..."

Cold memory (<0.3):

> "I vaguely recall... it might have been..."

Confidence reflects freshness. The system admits uncertainty.

Why This Matters

Current LLMs confabulate confidently.

Wheeler Memory lets AI say **"I'm not sure"** and **mean it**.

Grounded in theory (Wheeler's information dynamics).
Validated in practice (10K+ test cases).

Not Just Search

Vector DBs (traditional): Find stored text that matches the query.

- Retrieval-based
- Exact-ish match
- No reconstruction

Wheeler Memory: Find the closest pattern, blend it with your context, re-evolve it.

- Reconstruction-based
- Context-colored
- Meaning emerges from dynamics

Two Modes: Exact + Fuzzy Recall

Exact: Store "buy milk". Query "buy milk". Get the exact memory back.

Fuzzy: Store "grocery list". Query "shopping". Get a context-colored reconstruction.

Same system. Two recall philosophies.

The Philosophical Foundation

Wheeler: "**It from Bit**" — Information emerges from dynamics, not storage.

A memory isn't retrieved. It's **computed**. Reconstructed. Born anew each time.

Like you. Every recall rewrites the memory.

Design Philosophy: Engine ≠ Voice

Engine: The cellular automaton (the mind).
Voice: The LLM wrapper (the speaker).

Swap the LLM. The memory system stays the same.
The behavior persists.

Meaning comes from structure, not labels.

Privacy & Ownership

All local. No cloud APIs. No third-party servers.

Your memories stay on your machine.

Your data. Your automaton. Your voice.

Real Numbers

- **10K+** test cases across 6 domains
- **95%+** paraphrase coverage (semantic recall)
- **~3ms** CPU (convergence)
- **GPU**: 10× faster (HIP/ROCm + CUDA)
- **19 modules**, 167 tests, fully open-source

Validated. Tested. Ready.

What's Running Today

✓ Web UI dashboard

✓ CLI tools (store, recall, temps, scrub)

✓ Darman chatbot agent

✓ Sleep consolidation (spreading activation)

✓ Semantic embeddings (optional)

✓ GPU auto-fallback

Everything works locally. Nothing leaves your machine.

Live Demo

[Walkthrough of Web UI → Darman agent → CLI tools → under the hood]

Watch memories being stored, recalled, and reconstructed in real time.

What's Next

- **Images & Audio** — Extend to multimodal memories
- **Concept Clustering** — Organize related memories
- **Federated Memory** — Distributed learning between trusted peers
- **WebAssembly** — Run in the browser

The foundation is proven. Now we build.

Open Source

GitHub: fantomx42/wheeler-memory

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Community: Contributions welcome.

Why This Excites Us

First-of-a-kind memory architecture for AI.

Rooted in cellular automata theory.
Validated in practice.
Grounded in cognitive science (Elizabeth Loftus on reconstructive memory).

AI with a mind that remembers like you do.

The Formula Is The Foundation

Everything else is commentary.

Wheeler Memory: Not a search engine.
Not a retrieval system.
A **reconstruction engine**.

A mind. An automaton. A voice.

Darman remembers.